



THE UNIVERSITY OF
SYDNEY

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Phase 1: Project Proposal

Detecting Energy Poverty Hotspots in NSW

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1 Abstract

This project aims to identify energy poverty hotspots in New South Wales suburbs by analyzing deviations between predicted and actual energy consumption during extreme temperatures. Using machine learning models (Random Forest, LightGBM, XGBoost), daily energy usage and climate data are transformed into predictive insights, with residuals highlighting under-consumption indicative of affordability issues. Feature engineering and dimensionality reduction (PCA) refine drivers of demand. The approach balances computational efficiency with actionable social impact, offering a scalable framework to address energy poverty.

2 Project Justifications

Each year in Australia, millions of households struggle to afford their gas and electricity bills during peak winter and summer periods, exposing them to cold- and heat-related health risks. In fact, about one in four homes lives in what's known as energy poverty [1][2]. Reducing this figure would deliver substantial community benefits, such as:

1. Reduced energy poverty and health improvements
2. Increase in the economy and general productivity of people
3. Uphold the world with equity and inclusion, with no agendas or schemes

This method, compared with other existing methods such as Concession Rebates and Vouchers Manual Assessments and Surveys is better in such ways:

- **Impartiality:** This method doesn't require data on political affiliations, ethnicity, age, or residency status, relying solely on the neutral technical indicators of inability to afford electricity bills while avoiding bias [3].
- **Scalability and Adaptability:** Because this method relies on machine learning, it is absolutely possible to refine the method by adding more features or integrate with more advanced system or model in the future.
- **Efficiency:** This method is much faster and more efficient compared to traditional eligibility verification due to machine learning [4].

The cost to implement and improve this method would include the actual subsidize by the government for the households with energy poverty, data infrastructure and development for ML solution such as smart-meter streams, weather APIs, ML platform setup, and data scientist salaries [5].

3 Project Objectives

The overarching goal of this project is to identify suburbs in NSW experiencing energy poverty by analyzing their energy usage patterns in relation to local climate conditions. The subgoals are shown below in *Table 1*

Objective	Feature Variables (X)	Predicted Variable (Y)
Forecast energy needs under climate stress	Maximum temperature, Minimum Temperature, Day, Month, Solar exposure	Daily energy usage & Baseline energy consumption under specific climate conditions
Determine suburbs with energy poverty condition	Baseline and actual energy usage consumption under extreme climate conditions	List of suburbs as energy poverty hotspots based on negative residuals

Table 1: Project Objectives

4 Proposed Method of Solution

The project will follow three streamlined steps: Data Preparation and Feature Engineering (Week 9), Model Testing and Validation (Week 10-11), and lastly is Evaluation and Analysis (Week 12-13) indicated by *Figure 1*. During Data Preparation and Feature engineering, Energy usage data (15-minute intervals) is aggregated into daily totals per suburb and merged with climate variables (temperature, humidity). Categorical data is encoded using ordinal or one-hot encoding. To reduce dimensionality, PCA may be applied to correlated climate features.

Table 2 shows the proposed model that will be used to analyze the data. Random Forest, LightGBM, and XGBoost are trained on the processed dataset using time-series cross-validation to preserve temporal order. Models are evaluated on their ability to predict daily energy use, with a focus on capturing non-linear climate-energy relationships. Simpler models such as Linear and Ridge regression are not mentioned because of the linearity, but they will be tested as well to be compared with the nonlinear models.

Models are evaluated using metrics such as R^2 , MAE, and more, with residual analysis during extreme temperatures, to identify suburbs where usage deviates significantly from predictions. Feature importance (SHAP) and PCA variance retention are analyzed to confirm key drivers, and hotspots are finalized via cross-model consensus and severity thresholds.

Model	Key Features	Pros	Cons	Suitability
Random Forest	Ensemble of decision trees; handles non-linearities and interactions	No need for feature scaling; robust to outliers	Slower inference; may overfit without tuning	High (good for temp-energy non-linearities)
LightGBM	Trees with leaf-wise growth; optimized for speed and scale	Handles large datasets efficiently; automatic feature importance	Requires careful hyperparameter tuning	High (ideal for temporal + climate data)
XGBoost	Trees with depth-wise growth; built-in regularization	High performance; robust to overfitting	Slightly slower than LightGBM	High (competitive alternative to LightGBM)

Table 2: Comparison of Candidate Models

5 Ethical or Moral Consequences

When working with our data, it is important to consider the following ethical implications:

1. Bias & Fairness: The disclaimer on the Ausgrid website states that metering inaccuracies and errors may cause abnormalities in the data. This may lead to biased predictions.
2. Privacy: Although always a factor for data analysis, we are not collecting any data concerning the person's sensitive information (political affiliations, race, age, etc.), so the privacy of

this sort of data is not an issue to be concerned about. However, energy consumption patterns may inadvertently expose people's lifestyles, habits, or presence at home. Even so, since our data is based off of 'zones' rather than individual houses, privacy is still not a large concern.

3. Transparency: Transparency through the form of the detailed documentation of our method helps us gain public trust as well as allows for the model to be properly improved over time [6].

6 Data Acquisition

We source our data from Ausgrid and Board of Meteorology (BOM). Ausgrid provide information on how much power is delivered from substation to a suburb while BOM provide the climate data including weather, temperature, solar exposure and rainfall [7][8]. We aim to choose a suburb in Sydney with significantly low household income, which we have yet to identify its data. Both Ausgrid and BOM provide comprehensive data for all suburb in Sydney. Hence, deciding one currently is not a matter of concern.

Ausgrid's dataset comprises daily energy consumption, every 15 minutes at a specific suburb annually in a single CSV file containing 367 rows. We will be extracting daily peak value and adding row values together to obtain total energy usage. The CSV file from BOM contains 366 rows for each feature.

The blank entries in the datasets are due to instrumentation error from both Ausgrid and BOM sides. These empty data will be dropped or filled using interpolation during data cleaning process.

7 Team Composition and Organization

Table 3 displays the members of the group along with their respective strengths and task allocation for this project.

Member	Skills	Work Allocation
Dawod	Inept in data analytics and Machine Learning algorithms, including Python. With abilities in communication and presentation	Report writing and formatting, Research, ML implementation, and Presentation
Max	Data cleaning in python, communication, public speaking	Data cleaning, report writing, Presentation
Imran	Team player, reasoning skills	Project research, analyzing the model, Implementing the model
Phoebe	Skills in writing, Detail-orientated, Skills in presentation	Report writing, research, ML implementation, and Presentation

Table 3: Team Composition and Work Allocation

References

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8 Appendix

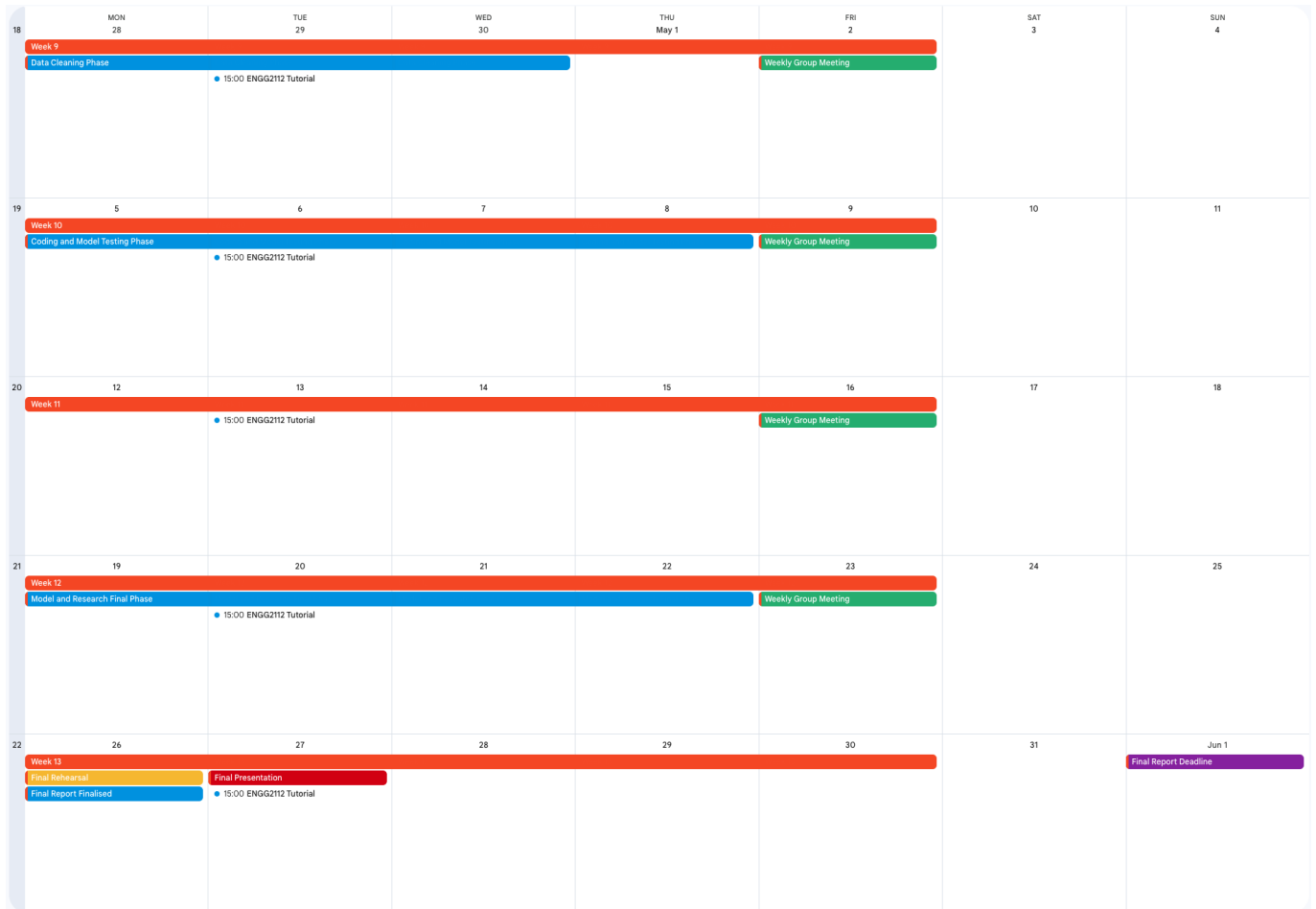


Figure 1: Group Timeline

ChatGPT were used in the process of making this proposal. The usage were limited to helping in research including paraphrasing conclusions, brainstorm ideas, and general questions. All answers provided by ChatGPT that ended up in the report were modified and refined.